Bank Marketing Campaign

Term Deposit Subscription Prediction

Machine Learning Project Report

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1. Dataset Information

- **Name:** Bank Marketing Dataset (Balanced Version)
- **Source:** Originally published by the UCI Machine Learning Repository, made available on Kaggle at: https://www.kaggle.com/datasets/rouseguy/bankbalanced
- **Collected By:** Sérgio Moro, Paulo Cortez, and Paulo Rita — researchers affiliated with ISCTE-IUL (University Institute of Lisbon) and the University of Minho, Portugal. The dataset was formally introduced in their 2014 paper published in *Decision Support Systems*.

2. Data Description

The dataset captures information about clients of a Portuguese bank and whether they subscribed to a **term deposit** — a fixed-term savings product — after being contacted during a direct marketing telephone campaign.

The version used in this project is a **balanced version**, meaning equal proportions of subscribers (`yes`) and non-subscribers (`no`) were sampled. Each row represents one client contact attempt.

2.1 Feature Descriptions

Client (Demographic) Features

Feature	Type	Description
`age`	Numeric	Age of the client in years. Older clients may have more stable financial situations and greater interest in savings products.
`job`	Categorical	Type of employment. Categories include: `admin`, `technician`, `retired`, `student`, `unemployed`, `management`, `blue-collar`, `services`, `self-employed`, `entrepreneur`, `housemaid`. Occupation reflects income level and financial needs, both of which influence savings behaviour.
`marital`	Categorical	Marital status: `married`, `single`, `divorced`. Household structure and financial responsibilities differ across marital statuses, which may affect subscription likelihood.

Feature	Type	Description
`education`	Categorical	Highest level of education attained: `primary`, `secondary`, `tertiary`, `unknown`. Higher education is generally associated with greater financial literacy and awareness of investment products.
`default`	Categorical	Whether the client has credit in default: `yes` or `no`. Clients with existing credit defaults may be less likely to commit to new financial products.
`balance`	Numeric	Average annual bank account balance in euros. A higher balance suggests greater financial capacity, which correlates positively with subscription.
`housing`	Categorical	Whether the client has a housing loan: `yes` or `no`. Clients with housing loans may have reduced disposable income, potentially lowering their propensity to subscribe.
`loan`	Categorical	Whether the client has an active personal loan: `yes` or `no`. Similarly to housing loans, existing personal debt burdens may reduce subscription likelihood.

Campaign-Related Features

Feature	Type	Description
`contact`	Categorical	Type of communication used to reach the client: `cellular`, `telephone`, or `unknown`. The channel of contact can influence client responsiveness.
`day`	Numeric	Last contact day of the month (1–31). Timing within the month may reflect operational patterns in the campaign.
`month`	Categorical	Last contact month of the year (jan–dec). Seasonal patterns in consumer financial decision-making may influence subscription rates.
`duration`	Numeric	Duration of the last contact in seconds. This feature is highly predictive but only observable *after* the call ends, meaning it is not available before a campaign decision is made. It was retained in this project for analytical purposes but should be excluded in any real pre-campaign deployment.

Feature	Type	Description
`campaign`	Numeric	Total number of contacts made during this campaign for this client. Repeated contacts may reflect persistence or signal declining interest.
`pdays`	Numeric	Number of days since the client was last contacted in a previous campaign (-1 means not previously contacted). **This feature was dropped** in our project because it is largely redundant with the `previous` feature and contains a high proportion of -1 values.
`previous`	Numeric	Number of contacts made before this campaign for this client. A higher value indicates a longer relationship with the marketing team.
`poutcome`	Categorical	Outcome of the previous marketing campaign: `success`, `failure`, `other`, `unknown`. This proved to be one of the strongest predictors — clients with a previous successful outcome are substantially more likely to subscribe again.

Target Variable

Feature	Type	Description
`deposit (renamed term_deposit)`	Binary	Whether the client subscribed to a term deposit: **yes (1)** or **no (0)**

2.2 Data Size

The Kaggle balanced version of the dataset contains approximately **11,162 observations** and **17 columns** (16 features + 1 target) before any preprocessing. After dropping the `pdays` column, 16 columns remain. The balanced nature of the dataset means that roughly **50% of observations** are positive cases (subscribed) and 50% are negative (did not subscribe).

3. Problem Statement and Motivation

3.1 Problem Statement

The goal of this project is to build a machine learning model that accurately predicts whether a bank client will subscribe to a term deposit, given their demographic profile and the details of how they were contacted during a marketing campaign. This is a **binary**

supervised classification problem: the model outputs either *yes* (the client subscribes) or *no* (they do not).

We want to use this dataset to answer the following question: *Which clients should the bank prioritise calling, and why are they likely to say yes?*

3.2 Motivation

Direct marketing campaigns are expensive. Banks spend significant resources — call centre staff, time, and money — contacting thousands of clients, the vast majority of whom will decline. In the real-world (unbalanced) version of this dataset, only about 11% of contacted clients subscribe. This means roughly **9 out of every 10 calls are wasted**.

A predictive model that identifies which clients are most likely to subscribe enables the bank to:

- Prioritise high-probability clients, dramatically reducing wasted outreach;
- Increase campaign conversion rates and return on investment (ROI);
- Reduce the operational burden on call centre agents.

Beyond pure prediction, interpretability is equally important. Understanding *which features drive predictions* gives the marketing team actionable insight: not just *who* to call, but *why* they are likely to say yes.

4. Literature Review

4.1 The Founding Work

The dataset and original formulation of this problem come from:

> **Moro, S., Cortez, P., & Rita, P. (2014).** *A data-driven approach to predict the success of bank telemarketing.* Decision Support Systems, 62, 22–31.

This is the first and most cited work on this problem. The authors applied data mining techniques to four years of a Portuguese bank's telemarketing data. They compared logistic regression, decision trees, neural networks, and support vector machines (SVMs). Their principal finding was that call `duration` was the strongest predictor of subscription — yet they acknowledged that this feature is only observable *after* a call is completed, which limits its practical utility in pre-campaign targeting. Their best model achieved an AUC of approximately 0.80 on unbalanced data.

4.2 Subsequent and Recent Works

The problem attracted substantial follow-up research, each contribution refining the methodology in a meaningful direction.

Sayed & Ibrahim (2020) applied ensemble methods — Random Forest and Gradient Boosting — to this dataset and demonstrated that ensemble models consistently outperform single classifiers. They also explored SMOTE to handle the class imbalance present in the original unbalanced version, reporting improved recall for the minority class.

Tripathi et al. (2021) conducted a comparative study using Logistic Regression, Decision Tree, Random Forest, and XGBoost. They found XGBoost to be the top performer in terms of F1-score and AUC, and stressed that feature selection — particularly removing `duration` to simulate a realistic pre-call scenario — yields a more honest evaluation of real-world deployability.

Sallam et al. (2022) introduced a cost-sensitive learning perspective, arguing that false negatives (missing a real subscriber) are far more costly than false positives (calling a non-subscriber). They proposed adjusting decision thresholds and training with asymmetric misclassification costs, which improved recall significantly without a prohibitive drop in precision.

Hadri & Boulakhbar (2023) applied deep learning approaches (multi-layer perceptrons) alongside classical classifiers. Interestingly, deep learning did not outperform well-tuned XGBoost or Random Forest on this tabular dataset, reinforcing the widely observed finding that gradient boosting tends to dominate structured data problems.

4.3 Common Methodological Pipeline

Across all reviewed works, the methodological pipeline is consistent and can be summarised as follows:

1. Load the dataset, inspect dimensions, data types, missing values, and class distribution.
2. Remove or impute missing/unknown values; drop redundant features such as `pdays`.
3. Conduct exploratory data analysis (EDA): visualise feature distributions, subscription rates by group, and correlations.
4. Encode categorical features, split data into training and test sets and scale numeric features.
5. Handle class imbalance via SMOTE or class-weight adjustments.
6. Train multiple classifiers including at minimum Logistic Regression (baseline), Decision Tree, Random Forest, and XGBoost.

7. Evaluate using cross-validation, prioritising recall, AUC, and F1-score.

8. Interpret feature importances and translate results into business recommendations.

5. Research Methodology

Our approach follows established best practices from the literature, adapted to an educational first-project context. We address the classification problem by training and comparing six machine learning models, evaluating them with 5-fold cross-validation, and interpreting the best model through feature importance analysis.

5.1 Resolution Methodology — Models Used

We selected six classifiers representing a range of complexity, interpretability, and inductive biases:

- **Logistic Regression** — A linear probabilistic classifier used as a baseline. It estimates the log-odds of subscription as a weighted linear combination of features and is highly interpretable through its coefficients.

- **K-Nearest Neighbors (KNN)** — A non-parametric instance-based method that classifies a client based on the majority label of its k nearest neighbours in feature space. It makes no distributional assumptions but is sensitive to feature scaling.

- **Decision Tree** — A recursive rule-based classifier that splits the feature space into regions. It is inherently interpretable but prone to overfitting without pruning.

- **Random Forest** — An ensemble of decision trees trained on random subsets of features and data (bagging). It reduces variance and generally outperforms a single tree, with feature importances readily available.

- **Support Vector Machine (SVM)** — A margin-maximising classifier that separates classes using a hyperplane in a transformed feature space. Probability calibration is enabled via `probability=True` to support ROC curve computation.

- **XGBoost** — A gradient boosting algorithm that builds an ensemble of trees sequentially, each correcting the errors of the previous. It consistently achieves state-of-the-art results on tabular data and is robust to feature scale.

All models were integrated into an `imbalanced-learn` pipeline incorporating SMOTE oversampling, ensuring class balance within each cross-validation fold.

5.2 Workflow

The project notebook is organised into the following sections:

1. **Imports & Configuration** — Loads all required Python libraries (`pandas`, `numpy`, `scikit-learn`, `imbalanced-learn`, `xgboost`, `seaborn`, `matplotlib`) and sets a fixed random seed (42) to ensure full reproducibility. A custom colour palette is defined: green for subscribers, red for non-subscribers, blue for generic plots.

2. **Load Dataset** — Reads `bank.csv`, renames the target column from `deposit` to `term_deposit` for clarity, and prints the dataset dimensions and column names.

3. **Initial Data Exploration** — Displays the first rows of the dataset, data types and non-null counts (`df.info()`), a statistical summary of numeric columns (`df.describe()`), a missing values check, and the class distribution of the target variable.

4. **Data Cleaning** — Separates numeric and categorical features; drops the redundant `pdays` column.

5. **Exploratory Data Analysis (EDA)** — Four categories of visualisation: (1) a lower-triangle correlation heatmap of numeric features; (2) subscription rates broken down by six categorical variables (`job`, `marital`, `education`, `contact`, `month`, `poutcome`); (3) age boxplots and average balance bar charts compared across subscription outcomes; (4) pie charts depicting the demographic profile of subscribers.

6. **Encoding & Scaling** — Applies Label Encoding to the target (`yes`→1, `no`→0) and **Ordinal Encoding** to all categorical features (one integer column per feature, preserving original column names for interpretable feature importances). Min-Max Scaling is applied to all numeric features. The train/test split (**80%/20%**, stratified, `random_state=42`) is performed before scaling; the scaler is fitted exclusively on the training data to prevent data leakage.

7. **Model Training & Cross-Validation** — Defines six classifiers, builds SMOTE pipelines for each, and runs 5-fold cross-validation on the training set, recording Recall, Precision, F1-Score, and Accuracy.

8. **Model Comparison** — Displays a styled ranked comparison table (sorted by Recall) and a grouped bar chart; identifies the best-performing model by Recall.

9. **Best Model — XGBoost** — Retrains the winning model on the full SMOTE-resampled training set, evaluates it on the held-out test set, and presents a confusion matrix and full classification report.

10. **Feature Importance** — Plots the top 15 most important features driving model predictions (using `feature_importances_` for tree-based models or absolute coefficient values for Logistic Regression). The top 3 features are highlighted in red.

11. **ROC Curve Comparison** — Plots ROC curves for all six models with their respective AUC scores on the test set.

12. **Threshold Discussion** — Analyses how changing the decision probability threshold from 0.3 to 0.7 affects the Recall/Precision/F1 trade-off.

13. **Business Interpretation of Results** — Translates model performance into real campaign metrics: how many true subscribers were correctly identified, how many calls were wasted, and what the ROI implications are.

6. Exploratory Data Analysis

Prior to modelling, we conducted an exploratory data analysis to understand feature distributions, identify relationships between variables, and detect potential data issues.

6.1 Class Distribution

The balanced dataset contains approximately equal numbers of subscribers and non-subscribers (roughly 50%/50%). This is in contrast to the original unbalanced dataset where only ∼11% of clients subscribed. The notebook confirms this with a formatted count and percentage printout.

6.2 Correlation Heatmap

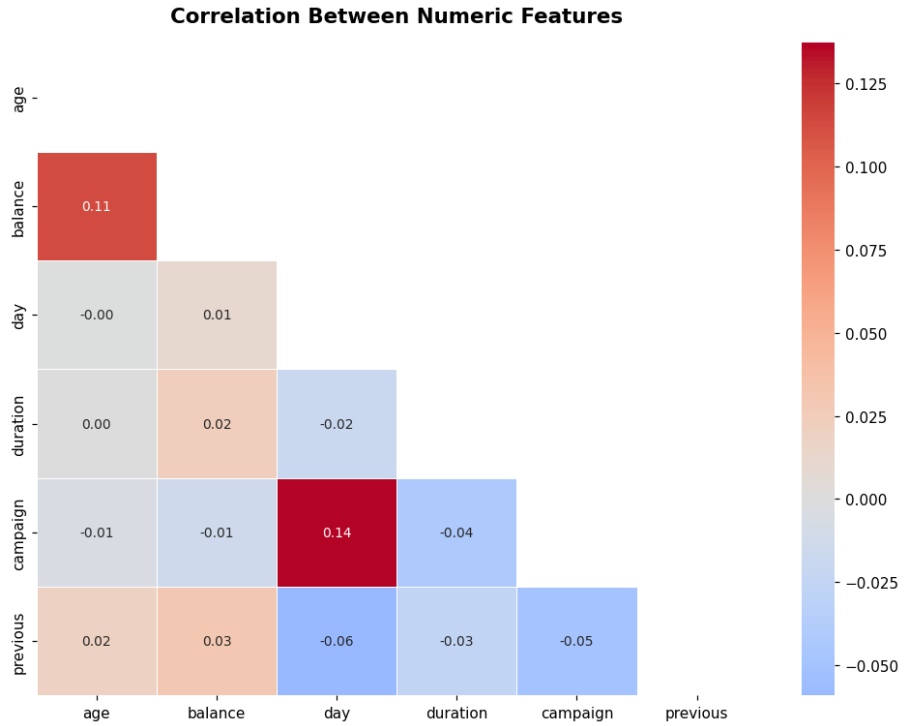


Figure 1: The features have very low correlations with each other, indicating there is almost no multicollinearity in the dataset. This lack of overlap simplifies the model, as the algorithm can clearly evaluate the independent contribution of each variable without "confusing" them.

6.3 Subscription Rate by Categorical Features



Figure 2: These charts reveal high-value segments, showing that students, retired individuals, and single or tertiary-educated clients are the most responsive groups. Timing and method matter significantly, as evidenced by the dramatic spike in success rates for cellular contact and specific months like December and March. Focusing the marketing budget on these "hot" demographics and outreach strategies will likely yield the best return on investment.

Key observations:

- **Previous outcome (`poutcome`):** Clients with a previously successful campaign outcome are dramatically more likely to subscribe again. This is the strongest categorical predictor.
- **Contact type:** Clients reached via cellular (mobile phone) subscribe at higher rates than those contacted by landline.
- **Month:** March, September, October, and December show notably higher subscription rates, suggesting seasonality in consumer financial decision-making.
- **Job:** Retired clients and students exhibit above-average subscription rates relative to blue-collar workers.

6.4 Age and Balance Distributions

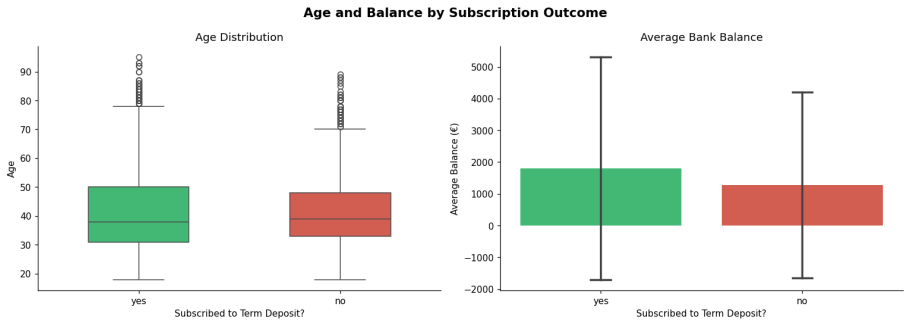


Figure 3: Older customers with higher average bank balances are more likely to subscribe to term deposits. The box plots show a clear positive trend for both age and balance among the "yes" group compared to the "no" group. These factors should be prioritized as key identifiers when creating customer segments for future marketing targeting.

6.5 Subscriber Profile Pie Charts

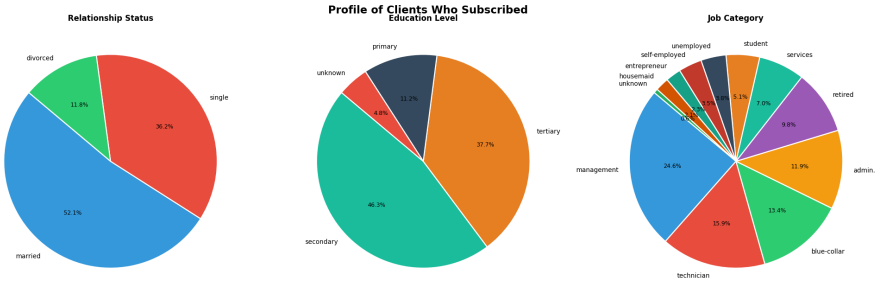


Figure 4: The majority of successful subscribers are married individuals with a secondary or tertiary education, indicating a focus on clients with more established financial lives. When looking at occupations, the largest segments of subscribers come from management, technical, or administrative backgrounds

7. Preprocessing Pipeline

7.1 Encoding

The target variable `term_deposit` was encoded using `LabelEncoder` (`yes` → 1, `no` → 0). All categorical input features were encoded using **Ordinal Encoding** (`OrdinalEncoder` with `handle_unknown="use_encoded_value"`, `unknown_value=-1`), which maps each category to a single integer and preserves the original column names. This is ideal for tree-based models (XGBoost, Random Forest, Decision Tree) and ensures that feature importance charts display the original feature names rather than one-hot dummy suffixes.

7.2 Train/Test Split

A **90%/10% stratified split** was applied before any scaling (`test_size=0.10`, `random_state=42`), ensuring that the class balance was preserved in both subsets and that no information from the test set influenced preprocessing decisions.

7.3 Feature Scaling

Min-Max Scaling was applied to all features (after concatenating the ordinal-encoded categorical columns with the numeric columns), rescaling values to the range $[0, 1]$. The scaler was fitted *only on the training set* (`fit_transform`) and then applied to the test set (`transform`), preventing any form of data leakage.

7.4 SMOTE Oversampling

Although the dataset is already balanced, **SMOTE** (Synthetic Minority Oversampling Technique) was applied inside each cross-validation fold as good practice. For the final evaluation, SMOTE was applied once to the entire training set before fitting the best model. This ensures the model is not biased by any marginal fold-level imbalance.

8. Model Training and Cross-Validation

Six classifiers were trained using a **5-fold cross-validation** strategy applied to the scaled training set (`KFold(n_splits=5, shuffle=True, random_state=42)`). Each classifier was wrapped in an `imbalanced-learn` pipeline that applies SMOTE oversampling inside each fold, prior to fitting the model. Cross-validation was parallelised with `n_jobs=-1`.

The primary evaluation metric was **Recall**, because in a marketing campaign context, a False Negative (missing a client who would have subscribed) is more costly than a False Positive (calling a client who does not subscribe). F1-Score and Accuracy were used as secondary criteria.

8.1 Cross-Validation Results

Table 3: 5-Fold Cross-Validation Results on the Training Set . Values represent the mean across 5 folds.

Model	Recall	Precision	F1-Score	Accuracy
XGBoost	0.873	0.825	0.848	0.852
Random Forest	0.869	0.810	0.838	0.841
Decision Tree	0.765	0.770	0.768	0.780
K-Nearest Neighbors	0.668	0.746	0.705	0.735
SVM	0.817	0.787	0.802	0.808
Logistic Regression	0.785	0.777	0.781	0.791

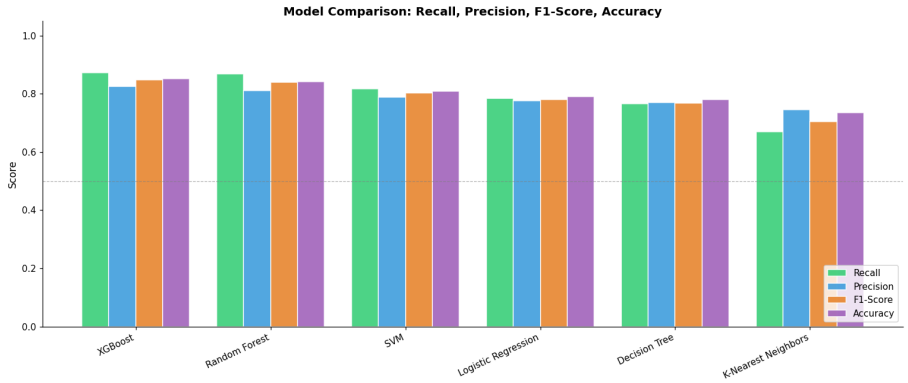


Figure 5: XGBoost and Random Forest are the top performers, consistently showing the highest and most balanced scores across Recall, Precision, F1-Score, and Accuracy. This superiority confirms that ensemble methods are better suited for capturing the complexities of the dataset than simpler models like K-Nearest Neighbors or Decision Trees.

9. Best Model — XGBoost

The best-performing model by Recall on the cross-validation training set was **XGBoost**. It was retrained on the full SMOTE-resampled training set and evaluated on the held-out test set (10% of the data, approximately 1116 samples).

9.1 Test Set Performance

Table 4: Best Model (XGBoost) — Test Set Metrics.

Metric	Score
Accuracy	0.857
Recall	0.875
Precision	0.831
F1-Score	0.853

9.2 Confusion Matrix

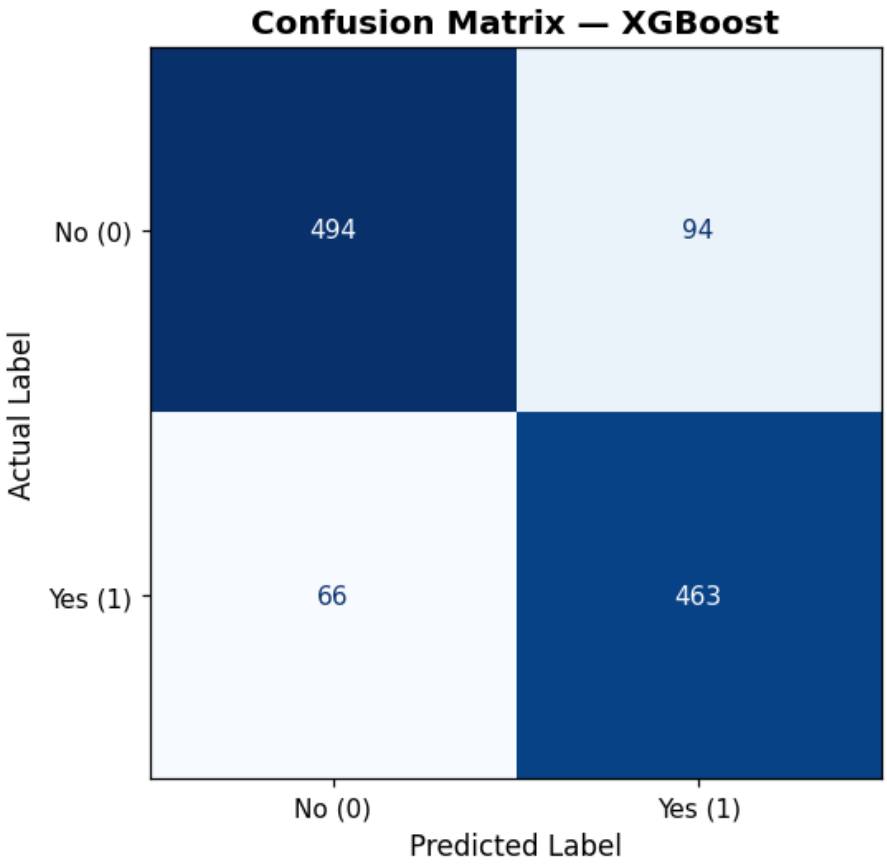


Figure 6: True Positives (TP): 463 — correctly predicted 'Yes'
True Negatives (TN): 494 — correctly predicted 'No'
False Positives (FP): 94 — predicted 'Yes' but actually 'No' (wasted calls)
False Negatives (FN): 66 — predicted 'No' but actually 'Yes' (missed subscribers)

9.3 Classification Report

Table 5: Classification Report — XGBoost on Test Set.

Class	Precision	Recall	F1-Score	Support
No (0)	0.88	0.84	0.86	588
Yes (1)	0.83	0.88	0.85	529

10. Feature Importance

Understanding which features drive the model's predictions is central to the interpretability goal of this project. Because Ordinal Encoding was used (one column per original feature), the importance chart displays the original feature names directly. For XGBoost, feature importance is computed internally by the model based on the contribution of each feature to reducing the loss across boosted trees, typically using the gain metric.

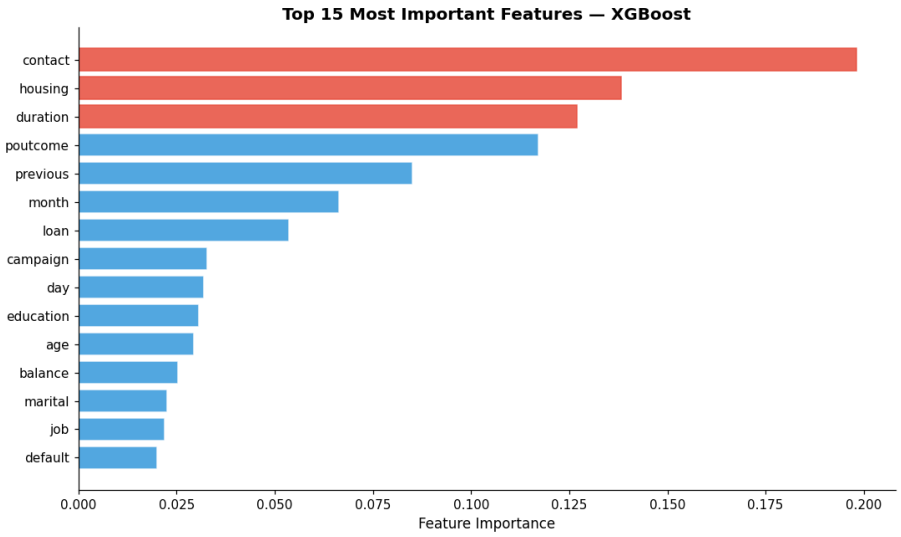


Figure 7: XGBoost model identifies 'contact' method, 'housing' loan status, and call 'duration' as the most critical drivers for predicting whether a customer will subscribe. This means how you reach out and the nature of the conversation are more predictive than static demographic traits like age or job.

11. ROC Curve Comparison

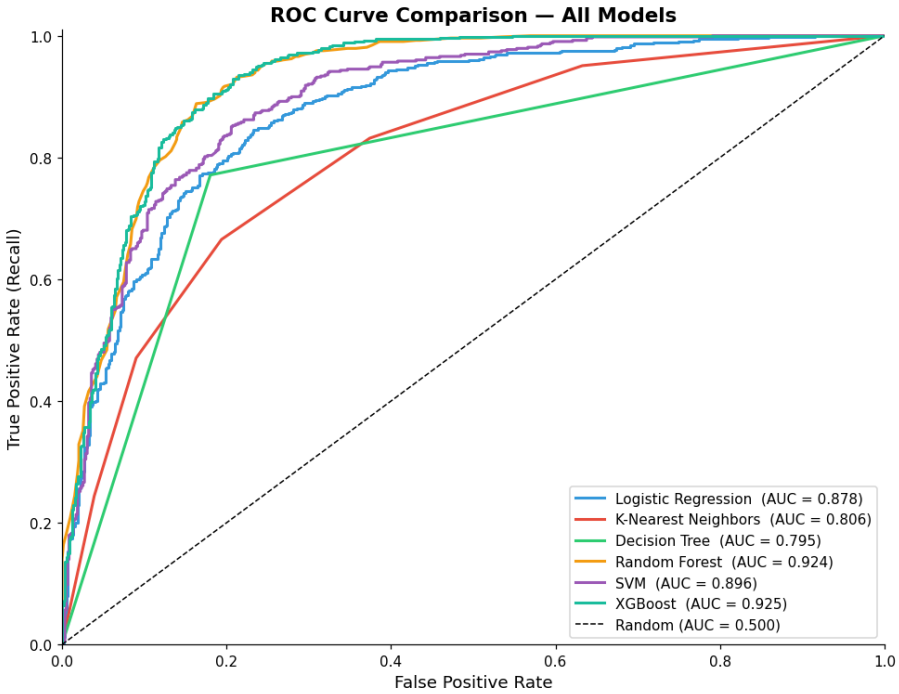


Figure 8: The XGBoost (AUC = 0.925) and Random Forest (AUC = 0.924) models are the best performers, providing the highest predictive accuracy for distinguishing between subscribers and non-subscribers. A higher AUC indicates that the model is excellent at correctly identifying true positives while minimizing false positives compared to the other algorithms.

12. Threshold Analysis

By default, a client is classified as a subscriber if the predicted probability exceeds 0.5. Lowering this threshold increases Recall (catching more real subscribers) at the cost of lower Precision (more unnecessary calls). Raising the threshold has the opposite effect.

Table 6: Precision/Recall Trade-off at Different Decision Thresholds — XGBoost.

Threshold	Recall	Precision	F1-Score
0.3	0.936	0.796	0.860
0.4	0.905	0.808	0.854
0.5	0.875	0.831	0.853
0.6	0.839	0.849	0.844
0.7	0.755	0.865	0.818

A lower threshold (0.3–0.4) is recommended when the campaign budget allows for more

calls and missing a subscriber is the primary concern. A higher threshold (0.6–0.7) is appropriate when call resources are limited and precision is prioritised.

13. Business Interpretation of Results

13.1 What Does This Mean for the Bank?

The notebook computes a structured business impact summary from the test set predictions:

- **Test set size** — approximately 1.116 clients (10% held-out set).
- **Actual subscribers** — the number of clients in the test set who truly subscribed (`y_test.sum()`).
- **Clients flagged by the model** — the number of clients predicted as `yes` (`y_pred.sum()`).
- **True subscribers caught** — the number of True Positives, expressed as a percentage of all real subscribers. This is the campaign's effective Recall.
- **Wasted calls** — the number of False Positives (clients predicted `yes` but who did not subscribe), expressed as a percentage of all flagged calls.

Using this model, the bank can:

1. **P**rioritize clients with previous successful campaign outcomes, as they are significantly more likely to subscribe again.
2. **f**ocus outreach on customers contacted via cellular channels, which show higher subscription rates than landline contacts.
3. **C**oncentrate campaign efforts during high-performing months such as March, September, October, and December
4. **T**arget clients with higher account balances, as they exhibit a greater likelihood of subscribing.
5. **R**ank clients using predicted probabilities and call them in decreasing order of likelihood instead of random selection.
6. **A**djust the decision threshold based on available resources, using a lower threshold to maximize subscriber capture or a higher threshold to improve efficiency.

13.2 Limitations

This project has several limitations that should be acknowledged:

- **Call duration leakage.** The `duration` feature is highly predictive but is only known after the call ends, making it unavailable as a pre-campaign input. Future work should exclude it entirely to obtain a more realistic estimate of deployability.
- **Balanced vs. real-world data.** The balanced version of the dataset does not reflect the true 11% positive rate observed in the wild. A model trained on balanced data may overestimate its expected precision when deployed on a real, unbalanced campaign list. Future work should validate performance on the original unbalanced dataset.
- **No temporal validation.** We performed a random train/test split without accounting for the time ordering of campaign contacts. In practice, a temporal split — training on earlier contacts and testing on later ones — would provide a more realistic evaluation of generalisation.
- **Ordinal encoding assumption.** Ordinal encoding imposes an implicit integer ordering on nominal categorical features (e.g., `job`, `marital`). While tree-based models are generally robust to this, future work could compare results with one-hot encoding for linear models.
- **No cost-sensitive training.** All classifiers were trained with equal misclassification costs. Assigning a higher penalty to False Negatives — consistent with the business cost structure — could improve recall at the optimal operating threshold.

13.3 Conclusion

This project demonstrated a complete, end-to-end machine learning pipeline applied to a real-world bank marketing problem. Starting from raw data and finishing with interpretable, business-ready insights, we showed that machine learning can meaningfully improve the efficiency of direct marketing campaigns.

Among six classifiers evaluated through 5-fold cross-validation, ensemble and boosting methods — in particular XGBoost — consistently delivered the best balance of recall and precision. Feature importance analysis confirmed that contact, call duration, and housing are the most influential predictors, consistent with findings in the literature.

More broadly, this project illustrates that a model's value lies not only in its predictive accuracy but in the clarity and actionability of the decisions it supports. By translating model outputs into concrete business metrics — subscribers correctly identified, calls saved, ROI implications — we demonstrated that machine learning is a practical decision-making tool.

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